Unit 1: Foundations and Prerequisites

Prepared by Staples Education

Condensed Cheat Sheet

Unit 1 at a Glance: Core Sub-Topics

This unit refreshes the calculus toolkit and the vocabulary the rest of the course assumes. It moves from derivatives and integrals, through the number e and Euler's formula, into the precise language of order, degree, linearity, and what it means to be a solution.

- **Calculus Review** — the power, product, and chain rules, and integrating the exponential.
- **The Number** e — the function that is its own derivative, and where it comes from.
- **Euler's Formula and Identity** — $e^{i\theta} = \cos\theta + i\sin\theta$ and the geometry of rotation.
- **Order, Degree, and Linearity** — the classification words that label every equation.
- **What It Means to Be a Solution** — verification by substitution, general versus particular.
- **Eliminating Arbitrary Constants** — building a differential equation from a family of curves.

Part I — Condensed Cheat Sheet

1.1 Calculus Review

Every solving technique ahead rests on fluent differentiation and integration. The three rules you reach for most are the power rule, the product rule, and the chain rule — and reversing them is exactly what integration asks of you.

Core Derivative and Integral Rules

$$\frac{d}{dx}x^n = nx^{n-1}, \qquad \frac{d}{dx}[fg] = f'g + fg', \qquad \frac{d}{dx}f\big(g(x)\big) = f'\big(g(x)\big)\,g'(x).$$

Integrating the exponential reverses the chain-rule factor:

$$\int e^{kx}\,dx = \frac{1}{k}e^{kx} + C.$$

Common Pitfall: The Missing Inner Derivative and the Lost $+C$

The chain rule multiplies by the derivative of the inside — $\frac{d}{dx}e^{3x} = 3e^{3x}$, not e^{3x}. And an indefinite integral always carries the arbitrary constant $+C$; dropping it discards the entire family of solutions an initial condition is meant to select from.

Worked Example 1.1: Differentiate two composites

(a) Chain rule on e^{3x}:

$$\frac{d}{dx}e^{3x} = 3e^{3x}.$$

(b) Power and chain rule on $(2x+1)^3$:

$$\frac{d}{dx}(2x+1)^3 = 3(2x+1)^2 \cdot 2 = 6(2x+1)^2.$$

1.2 The Number e

The natural exponential e^x is singled out by one property: it is its own derivative. That self-matching behavior is exactly what a rate proportional to the current amount demands, which is why e governs every growth and decay model in the course.

The Natural Exponential

$$\frac{d}{dx}e^x = e^x, \qquad \frac{d}{dt}e^{kt} = k\,e^{kt}.$$

The constant itself arises as a limit and as a series:

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \sum_{n=0}^{\infty}\frac{1}{n!} \approx 2.718.$$

Worked Example 1.2: Solve $\frac{dy}{dt} = y$

The equation asks for a function equal to its own derivative. The natural exponential matches, scaled by a free constant:

$$y = Ce^t, \qquad \frac{dy}{dt} = Ce^t = y. \quad \checkmark$$

The constant C supplies the whole family of solutions.

1.3 Euler's Formula and Identity

Euler's formula ties the exponential to rotation: feeding an imaginary number into e^x traces the unit circle. This is the bridge that lets a single complex exponential stand in for an oscillating sine–cosine pair.

Euler's Formula, Identity, and Polar Form

$$e^{i\theta} = \cos\theta + i\sin\theta, \qquad e^{i\pi} + 1 = 0.$$

A complex number in polar form is $z = re^{i\theta}$, where $r = |z|$ is the distance from the origin. Every point on the unit circle has $\left|e^{i\theta}\right| = 1$, and products multiply moduli while adding angles.

Worked Example 1.3: Evaluate by rotation

(a) A quarter-turn:

$$e^{i\pi/2} = \cos\tfrac{\pi}{2} + i\sin\tfrac{\pi}{2} = i.$$

(b) Multiply in polar form, multiplying moduli and adding angles:

$$\big(2e^{i\pi/6}\big)\big(3e^{i\pi/3}\big) = 6\,e^{i(\pi/6+\pi/3)} = 6\,e^{i\pi/2}.$$

1.4 Basic Definitions: Order, Degree, and Linearity

Before any technique applies, you classify the equation. The *order* is the highest derivative present; the *degree* is the power on that highest derivative once the equation is polynomial in its derivatives; and *linearity* asks whether the unknown and its derivatives appear only to the first power.

Classification Vocabulary

- **Order** — the highest derivative that appears.
- **Degree** — the power on the highest-order derivative.
- **Linear** — y and its derivatives appear to the first power only, never squared and never multiplied together.
- **ODE vs PDE** — one independent variable versus several.
- An n-th order equation needs n initial conditions to pin a unique solution.

Common Pitfall: Order Is Not Degree

Order and degree are different questions. In $\left(\dfrac{dy}{dx}\right)^3 + y = x$ the highest derivative is the *first*, so the order is one, while the cube makes the degree three. And $\dfrac{dy}{dx} + y^2 = 0$ is *nonlinear* — the squared y, not the derivative, breaks the rule.

Worked Example 1.4: Classify three equations

(a) $\dfrac{d^2y}{dx^2} + 3\dfrac{dy}{dx} = 0$ — order 2, degree 1, linear.

(b) $\left(\dfrac{dy}{dx}\right)^3 + y = x$ — order 1, degree 3, nonlinear.

(c) $\dfrac{d^3y}{dx^3} + \left(\dfrac{dy}{dx}\right)^2 = 0$ — order 3, degree 1 (the power sits on the first derivative, not the highest).

1.5 What It Means to Be a Solution

A solution is a function that makes the equation an identity — true for every x, not at one lucky point. You confirm a candidate by substituting it and its derivatives and checking both sides agree. The *general* solution keeps its arbitrary constants free; a *particular* solution fixes them.

Algorithm: Verifying a Candidate Solution

1. Compute every derivative of the candidate the equation requires.
2. Substitute the candidate and those derivatives into the equation.
3. Simplify the left and right sides independently.
4. Confirm the two sides are equal as an identity in x — for all inputs, not a single point.

Common Pitfall: "It Works at One Point"

Checking a candidate at $x = 0$ alone proves nothing — almost any function agrees somewhere. A genuine solution must satisfy the equation across its *whole* domain.

Worked Example 1.5: Verify two candidates

(a) Does $y = e^{2x}$ solve $y'' - 4y = 0$? Here $y'' = 4e^{2x}$ and $4y = 4e^{2x}$, so $y'' - 4y = 0$. ✓
(b) Does $y = \cos x$ solve $y'' + y = 0$? Here $y'' = -\cos x$, so $y'' + y = -\cos x + \cos x = 0$. ✓

1.6 Forming Differential Equations by Eliminating Constants

The reverse problem is just as instructive: given a family of curves with n arbitrary constants, you build the differential equation it satisfies by differentiating n times and eliminating the constants. The resulting order equals the number of independent constants.

Algorithm: Eliminating Arbitrary Constants

1. Count the independent arbitrary constants; call it n.
2. Differentiate the relation n times to produce n new equations.
3. Use those equations to solve for and eliminate every constant.
4. The result is an n-th order differential equation, free of constants.

Worked Example 1.6: Eliminate the constants

(a) From $y = Cx$: differentiate to get $y' = C$, and since $C = y/x$,

$$\frac{dy}{dx} = \frac{y}{x}.$$

(b) From $y = A\cos x + B\sin x$ (two constants): differentiate twice, $y'' = -A\cos x - B\sin x = -y$, so

$$\frac{d^2y}{dx^2} + y = 0.$$

Unit 1 Key Takeaway

This unit is the vocabulary and toolkit the whole course speaks in. Fluent differentiation and integration let you both check and build equations; the natural exponential e is the function equal to its own derivative; and the words order, degree, and linearity classify what you are looking at before any method is chosen. A solution is a function that satisfies the equation for all x, verified by substitution.